

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program – Sixth Semester – Spring 2025

Course Title: Artificial Intelligence

Course Code: D-326

Q.1: Answer the following short questions (6×5 = 30)

Q1. Suppose you have various sensors in a smart home system professionally used in the training set (e.g., motion sensors, temperature sensors, etc.). What challenges would you face when converting and improving data for use in a machine learning model? Explain the differences between supervised and unsupervised learning.

Answer:

Challenges in data conversion:

1. **Noisy data:** Sensors may give incorrect readings.
2. **Missing values:** Some sensors may fail or skip data.
3. **Different formats:** Data from different sensors may not match.
4. **Large volume:** Handling big data is difficult.
5. **Real-time processing:** Data must be processed quickly.
6. **Data imbalance:** Some events occur more than others.

Supervised vs Unsupervised Learning:

1. Supervised uses labeled data; unsupervised uses unlabeled data.
2. Supervised predicts output; unsupervised finds patterns.
3. Supervised examples: classification, regression.
4. Unsupervised examples: clustering, grouping.
5. Supervised needs training labels; unsupervised does not.

Q2. Write down a situation where a belief revision agent must revise its beliefs due to new evidence. How does it resolve inconsistencies?

Answer:

1. Example: A weather system predicts sunny weather.
2. New data shows dark clouds and rain signals.

3. The agent updates its belief from “sunny” to “rainy.”
4. It removes old conflicting information.
5. Uses priority rules or probability to choose correct belief.
6. Maintains consistency in knowledge base.

Q3. Without any numerical knowledge, how can we design a self-driving car system?

Answer:

1. Use rule-based decision systems.
2. Define conditions like “if obstacle → stop.”
3. Use symbolic reasoning for traffic rules.
4. Apply logical conditions for turning and braking.
5. Use maps and predefined paths.
6. Combine perception with decision rules.

Q4. Self-driving cars are working as a self-driving car system that must make decisions based on sensor data. How does it use a model-based reflex agent?

Answer:

1. It keeps track of environment using internal model.
2. Uses sensor data to update current state.
3. Predicts future situations using the model.
4. Takes actions based on current and past data.
5. Handles partially observable environments.
6. Improves decision-making over simple reflex agents.

Q5. Suppose you are working on a medical diagnosis system that must deal with uncertain and imprecise data. How can fuzzy logic be used effectively?

Answer:

1. Handles uncertainty using degrees (not true/false).
2. Uses linguistic variables like “high fever.”
3. Applies fuzzy rules for diagnosis.
4. Combines multiple symptoms smoothly.
5. Provides approximate but useful results.
6. Improves decision-making in unclear situations.

Q6. How game theory is related to everyday life?

Answer:

1. Helps in decision-making in competitive situations.
2. Used in business strategies.
3. Applies in negotiations and pricing.
4. Helps understand cooperation and conflict.
5. Used in politics and voting systems.
6. Improves strategic thinking in daily life.

Q.2: Answer the following questions ($3 \times 10 = 30$)

Q.2(a) Consider the given dataset with different examples, apply Naïve Bayes classification algorithm to predict whether a Car (Red, SUV, Domestic) is stolen or not.

Example No.	Color	Type	Origin	Stolen
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	SUV	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Domestic	Yes
8	Yellow	SUV	Imported	No
9	Red	SUV	Imported	No
10	Red	Sports	Domestic	Yes

Answer:

Step 1: Prior Probabilities

$$P(\text{Yes}) = 5/10 = 0.5$$

$$P(\text{No}) = 5/10 = 0.5$$

Step 2: Likelihood Probabilities

For **Yes (Stolen)**:

- $P(\text{Red}|\text{Yes}) = 3/5$
- $P(\text{SUV}|\text{Yes}) = 2/5$
- $P(\text{Domestic}|\text{Yes}) = 3/5$

For **No (Not Stolen)**:

- $P(\text{Red}|\text{No}) = 2/5$
- $P(\text{SUV}|\text{No}) = 3/5$
- $P(\text{Domestic}|\text{No}) = 2/5$

Step 3: Apply Naïve Bayes

For Yes:

$$P = 0.5 \times (3/5 \times 2/5 \times 3/5)$$

For No:

$$P = 0.5 \times (2/5 \times 3/5 \times 2/5)$$

Step 4: Comparison

Yes probability > No probability

Final Answer:

The car (Red, SUV, Domestic) is predicted as **Stolen (Yes)**

Q.2(b) Run A Search algorithm to calculate the optimal path from start node A to goal node J. Each node has an associated heuristic value $h(n)$ as follows:(Graph given in paper)*

Answer:

A Formula:*

$$f(n) = g(n) + h(n)$$

Where:

- $g(n)$ = path cost
- $h(n)$ = heuristic value

Steps:

1. Start from node A
2. Expand nodes with lowest $f(n)$
3. Keep track of visited nodes
4. Update path cost and heuristic
5. Continue until goal node J is reached

Path Selection:

- Compare all possible paths
- Choose path with minimum cost

Final Answer:

Optimal path: **A** → **C** → **G** → **J** (*example based on typical solution*)

Q.2(c) A digital marketing company operates 3 regional offices which serve. They own data centers and use cloud computing. The company aims to use AI-based solutions for marketing analytics and needs to choose between BFS and DFS algorithms. Answer the following:

- a) Suggest which algorithm could represent individual users in a graph?
- b) Suggest at least two suitable graph representation techniques.
- c) Compare BFS vs DFS in terms of efficiency.
- d) What if the company expanded more regions and employees?

Answer:

(a) Representation:

- Users can be represented as nodes
- Relationships as edges

(b) Graph Representation:

1. Adjacency Matrix
2. Adjacency List

(c) BFS vs DFS:

BFS:

- Explores level by level
- Uses queue
- Finds shortest path
- More memory usage

DFS:

- Goes deep first
- Uses stack
- Less memory
- May not find shortest path

(d) Expansion Impact:

1. Graph size increases
2. More nodes and edges
3. Higher computation required
4. Need efficient algorithms
5. Cloud resources help scaling